



Anand Narasimhan
Computer Science & Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 17/06/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	MVM Sr. Sec. School, Chetpet, Chennai	2021	98.20%
Matriculation	CBSE	MVM Sr. Sec. School, Chetpet, Chennai	2019	96.00%

Pursuing a **Honours** degree from the **CSE Department**, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 17** and was the **Tamil Nadu State Topper** in **JEE-Advanced** (2021)
- Secured **Rank 28** and **Rank 32** in the Indian National Olympiads in Physics and Chemistry respectively and attended the *Orientation cum Selection Camps* for both subjects (2021)
- Awarded the **KVPY Fellowship in Basic Sciences** twice by the Indian Government. Achieved by securing **National Rank 56** in the SX stream ('20) and **National Rank 99** in the SA stream ('19). (2019-20)
- Received an **AP Scholar with Distinction** award for scoring **5/5** in **7** different **AP** (Advanced Placement) exams in **Computer Science, Math, Physics, Chemistry and English Composition** (2017-2020)
- Achieved a **SAT** score of **1550/1600** with a perfect score of **800/800** in the **Math** section (2019)

RESEARCH EXPERIENCE

Approximating DFAs with RNNs | *Summer Research Internship* (Summer '23)

Guide : Prof. Eric Fabre, INRIA, Rennes

- Investigated the problem of approximating **DFAs** using **RNNs** trained on the corresponding regular language.
- Found significant clustering among the **RNN's hidden states**, similar to the discrete states of an automaton.
- Experimented with **Bi-RNNs**, **LSTMs** to improve clustering, **t-SNE** to plot higher dimensional states in 2D

Graph Hashing | *R&D Project* (January '24 - July '24)

Prof. Abir De, CSE, IIT Bombay

- Worked on designing a **Locality Sensitive Hashing** procedure on a similarity score between a query and a corpus document, represented as **graph embeddings**, to perform **subgraph matching** in the document.
- Analysed existing LSH mechanisms and **Fourier-transform** based hashing methods for set containment search.
- Modified the code for **ISONET**, a subgraph-matching based retrieval model, to incorporate our LSH mechanism

Machine Learning Intern | *Winter Internship* (Winter '22)

ZLabs, Zoho Corporation

- Implemented the **RewriteNet** model used in **Scene Text Editing** to produce **realistic synthetic data**
- Studied the problem of fitting **log-linear** models to data where the dependent variable can be **zero** or **negative** and implementing several estimators such as **iterative OLS (iOLS)**, **i2SLS**, **PPML** and **Multiplicative Poisson**

KEY PROJECTS

Fast Chat | *Course Project* (Autumn '22)

Instructor: Prof. Kavi Arya - Software Systems Lab

- Built a terminal-based **chatting software** using the python socket library and other open-source libraries
- Implemented functionalities such as **Group chats**, **sending images** and an **admin** functionality, with **end-to-end encryption** between users. Experimented with different load balancing strategies for multiple servers as well
- Maintained a database using **PostgreSQL** keeping track of user data, public keys and undelivered messages

Specformer - Spectral GNNs meet Transformers | *Course Project* (Autumn '23)

Instructor: Prof Abir De - Learning with Graphs

- Implemented the **Specformer** paper, which proposes a **Transformer**-based architecture to improve **Spectral GNNs**, applying self-attention on the set of eigenvalues of the Laplacian to learn better **Spectral Filters**
- Implementing **Deformed Attention** in order to **sparsify** the attention matrix, thus reducing the complexity

SCLP: A Language Processor for a Small C-like Language | *Course Project* (Spring '24)

Instructor: Prof. Uday Khedker - Compilers Lab

- Built a compiler that incrementally supported features in **C** including **conditionals**, **control flow** and **recursion**
- Generated intermediate representations - Abstract Syntax Trees, Three Address Codes, Register Transfer Language, MIPS assembly code with register allocation, using tools like **lex** and **yacc** for scanning and bottom-up parsing.

Operating Systems | *Course Project*

(Autumn '23)

Instructor : Purushottam Kulkarni - Operating Systems

- Extended memory management in xv6 by adding **on-demand paging** and **CoW** optimisation on shared pages
- Developed various synchronization primitives, like **semaphores** and **barriers**, utilising atomic instructions in xv6
- Added support for kernel space threads, a custom priority process scheduler and low level filesystem interaction

Railway Planner | *Course Project*

(Autumn '22)

Instructor: Prof. Supratik Charkaborty - Data Structures and Algorithms Lab

- Implemented various Algorithms (such as Quicksort, the KMP Algorithm and Graph Based Algos) and Data Structures (including Dictionaries, Hash-Maps, Binary Search Trees, AVL Trees, Heaps, Tries, Graphs etc)
- Built a Railway Planner with multiple features like **maintaining reviews**, **searching** for Journeys, **querying** for train stations, **sorting** reviews based on their rating and features like **auto-complete** while searching

Improving Prefetchers | *Course Project*

(Spring '23)

Instructor: Prof. Biswabandan Panda, Digital Logic Design and Computer Architecture

- Improved the performance of winning submission of 3rd Data Prefetching Championship on graph based traces.
- Tried to capture complex access patterns by using previously accessed program counters and page addresses.
- Incorporated other prefetching strategies like best-offset prefetching and achieved an improvement of about 10%.

Automatic Speech Recognition | *Course Project*

(Spring '24)

Instructor: Prof. Preethi Jyothi - Automatic Speech Recognition

- Utilized **Whisper** model for speech recognition task on Code-Switched Hinglish dataset, achieving **WER** of 0.493
- **Presented LatticeST**, a paper on incorporating Lattice inputs to purely **self-attentional models**
- Simulated a modular framework for global normalization using **WFSAs** for **Streaming Speech Recognition** task

Quantum Computing, ML and Quantum ML | *Seasons of Code*

(May 2022 - July 2022)

Web and Coding Club, IIT Bombay

- Implemented **Quantum Computing Algorithms** such as Quantum Teleportation, Deutsch-Jozsa, Quantum Fourier Transforms, Phase Estimation, Shor's Algorithm, Grover Search using the **Qiskit** SDK
- Modified a **Quantum General Adversarial Network**, coded in the **PennyLane** Python library with **PyTorch** support, that generates hand written images of zero, using the recently proposed **Patch Method** for the network

Manifold Representation Learning | *Course Project*

(Autumn '23)

Instructor: Prof. Preethi Jyothi - Artificial Intelligence and Machine Learning

- Exploring various approaches to non-linear dimensionality reduction using **Manifold Learning**, like **t-SNE**, **LLE**
- Analyzing and implementing a particular approach, **Markov-Lipschitz** Deep Learning, using **Markov Random Fields** and local **bi-Lipschitz** continuity to obtain better **local geometry preservation** and **robustness**

POSITIONS OF RESPONSIBILITIES

Manager | *Maths and Physics Club, IIT Bombay*

(May 2023 - Present)

- Leading a team of **8 people** in organising Math and Physics events for enthusiasts of the institute, such as quizzes, competitions, group discussions, trips and lab visits, while managing day-to-day logistics and social media pages

Teaching Assistant | *CSE and Math Departments, IIT Bombay*

(March 2023-Present)

- **CSE:** Handling labs, designing exams, creating additional course material and solving doubts of 200+ students in two courses, **Abstractions and Paradigms of Programming**, and **Introduction to Machine Learning**.
- **Mathematics:** Conducted weekly problem-solving sessions in Lin Alg and Calculus for a class of 50.

Mentor - Quantum Computing, Diving into ML | *Learners' Space*

(Dec 2022-July 2023)

Undergraduate Academic Council, IIT Bombay

- Guided **300+** students on a **Quantum Computing** course, providing weekly assignments and solving doubts.
- Mentored a group of **25 students** on the **basics of Machine Learning**, with a focus on Neural Networks

Class Representative | *CSE Batch of 2025 - Second Year*

(July 2022 - May 2023)

- Acting as the primary point of communication between the professors and a batch of 194 for all course logistics

RELEVANT COURSES UNDERTAKEN AT IITB

Mathematics

Real Analysis, Calculus I and Calculus II, Linear Algebra, Differential Equations

Computer Science

Learning with Graphs, Modern Information Retrieval, Discrete Structures, Data Analysis and Interpretation, Logic in CS, Algorithms, Extremal Combinatorics, Automata Theory, Machine Learning, Networks, Comp Architecture

Misc.

Quantum Information and Computing, Quantum Physics, Electricity and Magnetism

TECHNICAL SKILLS

Languages/Software

C/C++, Python, Java, Bash, Prolog, Git, L^AT_EX, Doxygen, Sphinx, AutoCAD

QC and Data Science

Qiskit, PennyLane, NumPy, Pandas, Matplotlib, Seaborn, TensorFlow, Keras, PyTorch

Web Dev & Databases

HTML, CSS, JavaScript, Bootstrap, SQL, Flask

EXTRACURRICULAR ACTIVITIES

- **Winner of Bazinga Math Quiz 2022** organised by Maths and Physics Club, IIT Bombay (2022)
- Participated in the **IIT Bombay Scrabble Open**, a **SAI rated** event of top-rated players. (2022-2023)
- Trained in *Carnatic Music, Vocals and Keyboard* and have given multiple performances (2010 - Present)